Why Radial Shell Gaps Do Not Control Nonnormal Riesz Projections

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Abstract

The shell-complete resonance atlas has positive radial gaps, yet its biorthogonal cloud projectors become extremely ill-conditioned. We prove that there is no contradiction. For

$$A_M = \begin{pmatrix} \lambda & M \\ 0 & \mu \end{pmatrix}, \qquad \lambda \neq \mu,$$

the spectral projector onto λ is

$$P_{\lambda,M} = \begin{pmatrix} 1 & M/(\lambda-\mu) \\ 0 & 0 \end{pmatrix},$$

so $\|P_{\lambda,M}\|_2 = \sqrt{1 + |M/(\lambda-\mu)|^2}$. The eigenvalue gap is fixed while the projector norm is unbounded.

In the RH-232 audit the minimum radial shell gap remains 7.40×10^{-5}, whereas the maximum projector norm reaches 2.26×10^{12} and the minimum gap-to-projector ratio is 6.40×10^{-16}. Log–log fits against the noise give descriptive growth exponents 7.41 and 6.55 in the two channels. Radial separation is therefore not a pseudospectral certificate. The result marks a forbidden shortcut but leaves projection-free determinant factorization open.

1 Spectral separation versus invariant-subspace separation

For normal operators a spectral gap controls the orthogonal spectral projection. For nonnormal operators, eigenvalue location and invariant-space angle are different data [1, 2]. RH-222 recorded the first; RH-232 measured the second [3, 4].

The simplest triangular family already contains the entire obstruction.

2 Exact fixed-gap counterexample

Theorem 2.1 (Fixed-gap projector blow-up)**.** *Let $\lambda \neq \mu$ and*

$$A_M = \begin{pmatrix} \lambda & M \\ 0 & \mu \end{pmatrix}.$$

The Riesz projector associated with λ is

$$P_{\lambda,M} = \frac{A_M - \mu I}{\lambda - \mu} = \begin{pmatrix} 1 & M/(\lambda-\mu) \\ 0 & 0 \end{pmatrix}.$$

Consequently

$$\|P_{\lambda,M}\|_2 = \sqrt{1 + \frac{|M|^2}{|\lambda-\mu|^2}} \longrightarrow \infty$$

as $|M| \to \infty$, while the eigenvalue gap $|\lambda - \mu|$ is unchanged.

Proof. The polynomial formula follows because the minimal polynomial is $(z-\lambda)(z-\mu)$. The displayed projector has only one nonzero row, so its operator norm is the Euclidean norm of that row. $\square$

Corollary 2.2. *No bound depending only on a positive eigenvalue gap, radial or otherwise, can control the norm of a nonnormal spectral projector.*

This statement does not say that gaps are useless. They identify a contour that avoids the spectrum. What they do not control is the resolvent on that contour, and the Riesz formula integrates the resolvent rather than the set of eigenvalues alone.

3 The missing datum is a resolvent bound

Theorem 3.1 (Contour-resolvent projector bound)**.** *Let Γ be a positively oriented rectifiable contour in the resolvent set of a bounded operator A, enclosing an isolated spectral component. Its Riesz projector satisfies*

$$P_\Gamma = \frac{1}{2\pi i}\int_\Gamma (zI-A)^{-1}\,dz, \qquad \|P_\Gamma\| \le \frac{\operatorname{length}(\Gamma)}{2\pi}\sup_{z\in\Gamma}\left\|(zI-A)^{-1}\right\|.$$

Consequently, a uniform contour length and a uniform resolvent bound do give a uniform projector bound; a spectral gap alone supplies neither.

Proof. The first identity is the Riesz functional calculus. Taking norms under the Bochner integral gives the second inequality. $\square$

For the triangular family the resolvent can be written explicitly:

$$(zI-A_M)^{-1} = \begin{pmatrix} (z-\lambda)^{-1} & M((z-\lambda)(z-\mu))^{-1} \\ 0 & (z-\mu)^{-1} \end{pmatrix}.$$

On the circle $|z-\lambda| = r < |\lambda-\mu|$, its norm is at least the modulus of the upper-right entry, and hence at some—indeed every—point obeys

$$\left\|(zI-A_M)^{-1}\right\| \ge \frac{|M|}{r(|\lambda-\mu|+r)}.$$

The contour remains a fixed positive distance from both eigenvalues while the resolvent diverges linearly in $|M|$. The same off-diagonal coupling that creates the oblique projector therefore appears as pseudospectral inflation.

Remark 3.2. If $A = V\Lambda V^{-1}$ is diagonalizable, the elementary estimate

$$\left\|(zI-A)^{-1}\right\| \le \frac{\|V\|\,\|V^{-1}\|}{\operatorname{dist}(z,\sigma(A))}$$

shows exactly which datum supplements the spectral gap: a controlled eigenbasis condition number. In infinite-dimensional transfer problems an adapted resolvent estimate is the more invariant formulation.

4 Archived dynamical audit

The finite model uses $\lambda = 0.8$, $\mu = 0.5$, and couplings from 1 to 10^9. The gap stays 0.3 while the projector norm grows by a factor 9.58×10^8.

For the RH-232 endpoint family, the projector norms grow much faster than the radial gaps shrink. Representative values are shown below.

The fitted power exponents are diagnostics, not asymptotic theorems: the log residuals are large and the rank changes with the noise. The robust conclusion is only that no bounded trend is visible and radial gaps alone cannot supply one.

σ	left $\|P\|_2$	right $\|P\|_2$	conclusion
0.04	4.16×10^1	4.07×10^1	already oblique
0.008	2.79×10^5	1.10×10^6	unstable
0.0032	4.34×10^9	3.62×10^9	severe
0.00125	2.26×10^{12}	5.44×10^{10}	extreme

Table 1: Projector norms inherited from RH-232.

5 Scope of the negative result

Corollary 2.2 is a no-go theorem for any argument whose only input is eigenvalue location. It does not prohibit estimates that use positivity, reversibility, a symmetrizing density, analytic distortion, or a semiclassical anisotropic norm. Nor does it prove that the archived projectors actually diverge in the continuum limit: the finite exponents are descriptive and the matrices change dimension with σ.

The practical distinction is useful. A future positive result must expose where its resolvent control enters; it cannot cite the shell gap as a proxy. Conversely, failure of the Euclidean projector does not contaminate spectral quantities, such as a canonical determinant product, that depend only on algebraic eigenvalues.

6 Route consequence

The RH-222 shell gaps remain valid finite spectral observations. What fails is the implication

$$\text{positive radial gap} \quad \Longrightarrow \quad \text{uniformly bounded Riesz projector.}$$

Any operator-level continuation must add a resolvent estimate, an adapted norm, or a different factorization principle. RH-234 uses the canonical eigenvalue product for $\det_2$, which evaluates a finite cloud factor without ever multiplying by the ill-conditioned projector.

This paper does not exclude a certified contour on a different Banach space and does not close Gate A or any later arithmetic gate.

References

[1] Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, 1995.

[2] Lloyd N. Trefethen and Mark Embree. *Spectra and Pseudospectra*. Princeton University Press, 2005.

[3] Liang Wang. A rank-growing conjugate resonance-cloud atlas. 2026. RH-222 preprint.

[4] Liang Wang. Biorthogonal riesz-cloud projection and its conditioning wall. 2026. RH-232 preprint.